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Glued laminated timber and steel beams

A comparative study of structural design, economic and environmental consequences

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Abstract

Purpose – This paper aims to compare glued laminated timber and steel beams with respect to structural design, manufacturing and assembly costs and the amount of greenhouse gas emissions.

Design/methodology/approach – This paper presents structural design requirements in conformance with EN 1993: Eurocode 5 and Eurocode 3. With the help of these standards, expressions are derived to evaluate the design criteria of the beams. Based on the results of life-cycle analysis, the economic properties and environmental impact of the two types of beam are investigated. In this paper, the effect of beam span on the design values, costs and carbon dioxide emissions is analysed when investigating aspects of the structural design, economy and environmental impact. Different cross-sections are chosen for this purpose.

Findings – The study shows that the glued laminated (abbreviated as "glulam") beams have a smaller tendency to lateral torsional buckling than the steel beams, and that they can be cheaper. From an environmental point of view, glulam beams are the more environmentally friendly option of the two beam materials. Furthermore, glulam beams may have a direct positive effect on the environment, considering the carbon storage capacity of the wood. The disadvantage of glued wood is that larger dimensions are sometimes required.

Research limitations/implications – Wind load and the effect of second-order effects have not been considered when analysing the static design. Only straight beams have been studied. Furthermore, the dynamic design of the beams has not been investigated, and the bearing pressure capacity of the supports has not been analyzed. We have investigated timber beams with a rectangular cross-section, and steel beams of rolled I-sections, known as "HEA profiles". The cost analysis is based mainly on the manufacturing and assembly costs prevalent on the Swedish market. The only environmental impact investigated has been the emission of greenhouse gases. The design calculations are based on the European standards Eurocode 5 and Eurocode 3.

Practical implications – To achieve sustainability in construction engineering, it is important to study the environmental and economic consequences of the building elements. By combining these two effects with the technical design of buildings made of steel and/or timber, the concept of sustainable development can be achieved in the long run.

Social implications – The study concerns sustainability of building structures, which is an important of the sustainable development of the society.

Originality/value – The paper contains new information and will be useful to researchers and civil engineers.

Keywords Cost analysis, Eurocode 3, Eurocode 5, Greenhouse effects, Timber and steel beams

Paper type Research paper

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1. Introduction

Sustainable development has become a critical issue in long-term global development due to fossil fuel depletion, climate change and the increasing costs of energy and water. In this context, sustainable development aims in the first place to achieve economic development, social development and environmental protection.

Buildings have a huge impact on the environment, resource use, human health and productivity. For example, buildings account for 32 per cent of total energy consumption (Hassan, 2016a). When designing building constructions, it is necessary to consider economic development, social development and environmental development when aiming for sustainability. One of the key issues in this context is the choice of construction materials. This article considers two construction materials that are commonly used in building constructions, namely, wood and steel. It compares beams that are made of glued laminated timber (glulam beams) with steel beams, with respect to structural design, manufacturing costs, assembly costs and greenhouse gas emissions.

The construction and building sector accounted for 19 per cent of Sweden's carbon dioxide emissions in 2014, which corresponds to approximately 11.6 million tonnes of carbon dioxide (Boverket, 2017). The figure becomes 20.2 million tonnes of carbon dioxide when imported products are included. These figures show how important environmental engineering of building constructions is, to reduce its negative impact on the environment. The EU Commission and the Swedish Parliament have set a goal of climate neutrality by 2050, and to achieve this, it will be necessary to examine the construction sector.

It is important to consider also economic factors in construction, where the economic consequences of choice of materials can be critical.

Furthermore, the design properties of different construction materials differ due to the requirements made and safety criteria set for each type of building structure. The design must be able to carry safely the loads to which it is exposed. In addition, choice of material must ensure that the dimensions of each component are proportional to the final design.

Glulam (glued laminated timber) is a construction material that is composed a number of individual wood laminates bonded together with durable, moisture-resistant adhesives. The grain of all laminations runs parallel to the length of the member. In straight glulam products, the laminates are 45 mm thick. Glulam can be used in horizontal applications as a beam, or vertically as a column, and is available in a range of strength classes.

Steel beams are commonly made of structural steel, and steel members are usually manufactured with various standardised dimensions. Rolled steel sections are cast in continuous casting moulds without welded joints. Beams of rolled I-sections, known as "HEA profiles", are extensively used in the construction industry and are available in a variety of standard sizes. Such beams consist of two flanges with a web that connects them.

Few comparison studies between different construction materials with respect to structural design, economy and environmental impact have been published. Hassan (2003, 2004) studied wall and floor structures with respect to different criteria such as design, environmental impact and economy, using value-focused thinking as a decision tool.

Petersen and Solberg (2002) compared timber beams and steel beams at Gardemoen Airport in Norway with respect to environmental impact and economy. Sandin *et al.* (2013) reviewed the life-cycle assessments of several construction materials for their impact on the environment and found that the assumptions made in end-of-life modelling are important in the final assessment. The environmental impact of wood has been compared with that of other building materials by Hill and Dibdiakova (2016). Ekvall (2006) summarized and compared the environmental impacts of different construction materials, with a focus on greenhouse gases. Cole (1999) studied the transport and logistic considerations for several materials and examined also support processes. Brunklaus and Baumann (2002) considered what an increase in wood construction would mean for Sweden's environment. Sandberg *et al.* (2001) investigated the environmental impact of the steel industry. Gustavsson *et al.* (2006) investigated the greenhouse gas mitigation due to the production of wood materials. With respect to only the dynamic analysis of steel and timber beams, Hassan and Girhammar (2013) compared timber and lightweight steel floors with respect to footfall-induced vibrations.

In the previous works (Hassan, 2003; Hassan, 2004; Petersen and Solberg, 2002; Sandin *et al.*, 2013; Hill and Dibdiakova, 2016; Ekvall, 2006; Cole, 1999; Brunklaus and Baumann, 2002; Sandberg *et al.*, 2001; Gustavsson *et al.*, 2006), a comparison between timber beams and steel with respect to the structural design has not specifically been considered. In Hassan and Girhammar (2013), only the dynamic aspect of the timber and steel constructions is investigated. Moreover, the integration of the economy and the environment together with the structural design to pave the way to sustainable solutions for the built environment has not been paid attention to. This paper attempts to bridge this gap by integrating the design criteria with the economic and environmental consequences. To achieve sustainability in construction engineering, it is important to study the environmental and economic consequences of the building elements. By combining these two effects with the technical design of buildings made of steel and/or timber, the concept of sustainable development can be achieved in the long run.

The purpose of this study is to provide designers with an insight into the consequences of choosing either timber beams of rectangular cross-sections or steel beams of HEA profile (rolled I-sections) in roof constructions. It examines this choice from structural, economic and environmental perspectives. Static analysis is used to analyze structural design, while dynamic design is not investigated as it has been previously discussed (Hassan and Girhammar, 2013). Furthermore, this article attempts to provide an insight into the environmental impact of the two types of beam in terms of carbon dioxide equivalents emitted during the lifetime of the materials. This study examines the following questions:

Q1. What structural design criteria are important when choosing between steel and timber beams in typical roof constructions?

Q2. What are the financial consequences of choosing each material?

Q3. How large are the emissions of greenhouse gases from the two materials?

2. Methodology

The structural design requirements of the two beam types are based on the European standards EN 1993: Eurocode 5 and Eurocode 3. For the environmental assessment, the impact on the environment with respect to the emission of carbon dioxide is considered. The environmental data are based on the findings of previous research on the life-cycle analysis of the two beam types, as discussed later. For the economy, the material cost and "ready to assemble" cost for each beam type have been used in the assessment, based on the Swedish construction market.

The comparison methodology is to analyse the effect of beam span on the design values, costs and carbon dioxide emissions. Different cross-sections are chosen for this purpose. It is thought that using different beam spans is more feasible than to only compare different cross-sections of the two beams (timber and steel). This is because, in practice, the geometries of cross-sections of the two beams are not the same. In timber beams, rectangular cross-sections are usually used. In steel beams, the cross-sections often have the form of I-sections. Technically, the beam span influences the cross-sections as longer spans will require bigger dimensions if the structural requirement has to be satisfied; see e.g., Tables I to III, which show some results based on Section 3.

Further details on the design for the aspects of structural design, economy and environment of each beam type are presented in the following sections.

3. Structural design of steel and timber beams

The performance of structures should satisfy two basic requirements. The first is safety, usually expressed in terms of load-bearing capacity, which is referred to as the "ultimate limit state", and the second is serviceability, which describes the ability of the structural system and its elements to perform satisfactorily in normal use (Handbook 2, 2008).

The loading on the beam is chosen to represent action values, which are in most cases the values that the roof beams will experience in practice.

3.1 Structural model

For the analysis, the structural member is considered to be a simply supported beam as shown in Figure 1. This structural model represents roof beams that carry roof slabs. The loading actions on the beam are the characteristic permanent action: the self-weight of the roof slab (including covering materials and partitions) and the characteristic value of snow action on the roof. The value of snow action is taken to represent an extreme weather condition applicable in a northern Swedish town (Kiruna). Consequently, the loading actions

Span (m)	Steel section	Glulam cross-section: $b \times h$ (mm × mm), class GL30c
2	HEA100	215 × 180
4	HEA200	215 × 360
6	HEA260	215 × 540
8	HEA320	215 × 720
10	HEA450	215 × 945
12	HEA700	215 × 1305

Table I. Dimension of the steel and timber beams

Span (m)	$b \times h$	Moment check, Eq. (24)	% Reduction due to LTB, Eq. (25)	Shear check, Eq. (40)	Deflection check, Eq. (14)
2	215 × 180	0.856	0.0	0.541	0.613
4	215 × 360	0.900	0.0	0.897	0.647
6	215 × 540	0.943	0.0	0.903	0.681
8	215 × 720	0.993	0.0	0.910	0.715
10	215 × 945	0.996	8.0	0.864	0.661
12	215 × 1305	0.982	26.4	0.736	0.491

Table II. Results of structural analysis of the glulam timber beams, strength class GL30c

Span (m)	Steel profile	Moment check, Eq. (36)	%Reduction due to LTB, Eq. (37)	Shear check, Eq. (44)	Deflection check, Eq. (16)
2	HEA100	0.928	12.5	0.271	0.641
4	HEA200	0.756	18.8	0.229	0.489
6	HEA260	0.863	24.9	0.218	0.588
8	HEA320	0.957	31.3	0.205	0.641
10	HEA450	0.954	44.2	0.163	0.457
12	HEA700	0.985	63.1	0.112	0.238

Table III. Results of structural analysis of the steel beams calculated as their characteristic values are as follows: snow load, S = 25.92 kN/m, self-weight of roof slab, g_k = 2.4 kN/m. The self-weight of the timber beam (kN/m) is given by:

$$g_{k,\,beam} = \rho bh \cdot 9.82 \cdot 10^{-3} \tag{1}$$

where ρ is the volume density of the beam (kg/m^3), and h and b are the height (m) and width (m) of the beam, respectively. The self-weight of the steel beam with an HEA section (kN/m) is given by:

$$g_{k,\,beam} = g \cdot 9.82 \cdot 10^{-3} \tag{2}$$

where g is the beam weight (kg/m) as given in standard tables of rolled I-sections (Tibnor, 2007). The design value of the effects of actions q_{Ed} is determined by combining the values of actions that the designer believes may occur simultaneously. Only the main cases of strength (STR) of structures and structural members have been considered in this study, as it is thought to yield the design loading value.

The design value of action (q_{Ed}) may be calculated by considering the ultimate limit state and service limit state as shown below (Hassan, 2017).

Ultimate limit state

Load combination, STR-A:

$$q_{Ed} = \gamma_d 1.35 g_k + \sum_i \gamma_d 1.5 \psi_{0,i}\, q_{k,i} \tag{3}$$

Load combination, STR-B:

$$q_{Ed} = \gamma_d 1.2 g_k + \gamma_d 1.5 q_{k,1} + \sum_i \gamma_d 1.5 \psi_{0,i}\, q_{k,i} \tag{4}$$

Service limit state

Frequent load combination:

$$q_{Ed} = g_k + \psi_1 q_{k,1} + \sum_i \psi_{2,i}\, q_{k,i} \tag{5}$$

where ψ_0 is the factor for the combination value of a variable action, ψ_1 is the factor for the frequent value of a variable action, ψ_2 is the factor for the quasi-permanent value of a variable action and γ_d is a partial safety factor. Equation (5) is applicable for more severe conditions than the characteristic load combination, as it assesses the temporal damage or reversible state of the structure. Subsequently, it will be used to calculate the deflection in this study.

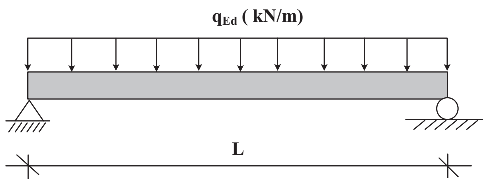


Figure 1.
Simply supported beam is loaded by the design load q_{Ed} and with span length L

For the beam in Figure 1, the design moment, M_{Ed}, and the design shear force, V_{Ed}, are given by:

$$M_{Ed} = \frac{q_{Ed}L^2}{8}\ ;\ V_{Ed} = \frac{q_{Ed}L}{2} \tag{6a, b}$$

3.2 Serviceability limit state

3.2.1 Glued laminated timber beam. The final deflection below a straight line between the supports, $u$fin, is (EN 1993 Eurocode 5, 2004):

$$u_{fin} = u_{fin,g} + u_{fin,q,1} \tag{7}$$

where $u_{fin,g}$ is the final deflection due to the permanent action:

$$u_{fin,g} = u_{inst,g}\left(1 + k_{def}\right) \tag{8}$$

and $u_{fin,q,1}$ is the final deflection due to the leading variable action:

$$u_{fin,q,1} = u_{inst\,g,1}\left(1 + \psi_{2,1}k_{def}\right) \tag{9}$$

where k_{def} is a deformation factor whose value depends on the type of material being stressed and on its moisture content, and u_{inst} is the instantaneous bending deflection, which may be written in this case in its general form:

$$u_{inst} = \frac{5q_{Ed}L^4}{384E_{0,\,mean}I} \tag{10}$$

where E_0, $_{mean}$ is the mean value of the modulus of elasticity parallel to the grain, I is the second moment of area of a section, $I = bh^3/12$, and L is the design span.

For thick beams ($L/h < 10$), it will be physically more realistic to include the shear deformations (Hassan, 2016b). These can be expressed in terms of the flexural deflection multiplied by a shear amplification factor, w_s (Porteous and Kermani, 2007):

$$w_s = \left(1 + 0.96\frac{E_{0,\,mean}}{G_{0,\,mean}}\left(\frac{h}{L}\right)^2\right) \tag{11}$$

Now if we set $k_{def} = 0.6$, $\psi_2 = 0.2$ and combine equation (7) with equation (11), we obtain:

$$w_{tot} = w_s(u_{fin,g} + u_{fin,q,1}) = w_s\left(1.12u_{inst,q,1} + 1.60u_{inst,g}\right) \tag{12}$$

Deflection checks should be made against unfactored permanent actions and unfactored variable actions. In this case, the load combination chosen is the frequent load combination, equation (5). The maximum deflection calculated must not exceed the deflection limit. In the case considered here, and considering a floor member with a plastered or plasterboard ceiling, the deflection limit, w_L is set to (EN 1993 Eurocode 5, 2004):

$$w_L = \frac{L}{250} \tag{13}$$

The deflection check can then be calculated as:

$$\frac{w_{tot}}{w_L} \leq 1.0 \tag{14}$$

Equation (14) is used here to check the deflection of the timber beam.

3.2.2 HEA beam. The maximum deflection is calculated as:

$$w_{tot} = \frac{5q_{Ed}L^4}{384EI} \tag{15}$$

where q_{Ed} is from the value given by equation (5), E is the elasticity modulus (210 GPa) and I is the second moment of area of a section in the strong axis, I_y (m^4). As $L/h > 10$ for the steel section chosen for the study, shear deformation may be neglected. The deflection limits for steel structures are not given directly in Eurocode 3, and thus the deflection limit has been set as determined by equation (13), to be able to compare results for the two materials. The deflection check is then:

$$\frac{w_{tot}}{w_L} \leq 1.0 \tag{16}$$

where w_{tot} is obtained from equation (15).

3.3 Ultimate limit state, moment check

3.3.1 Glued laminated timber beam. The most common use of a beam is to resist loads by bending about its major principal axis.

In the case in which a moment M_y only exists about the strong axis y, the stresses should satisfy (Handbook 2, 2008):

$$\sigma_{m,d} \leq k_{crit} f_{m,d} \tag{17}$$

where $\sigma m,\ d$ is the design bending stress, k_{crit} is a factor that takes into account the reduced bending strength due to lateral buckling and $f_{m,d}$ is the design bending strength. This is given by:

$$f_{m,d} = \frac{k_{mod} f_{m,k} k_h}{\gamma_M} \tag{18}$$

where k_{mod} is a modification factor that accounts for the effects of the duration of the load and the moisture content of the beam, $f_{m,k}$ is the characteristic value of bending moment capacity, and γM is the partial factor for a material property. For depths in the bending of glued laminated timber less than 600 mm, the characteristic values for $f_{m,k}$ may be increased by the factor k_h:

$$k_h = min\left\{\begin{array}{l}\left(\dfrac{600}{h}\right)^{0.1}\\ 1.1\end{array}\right. \tag{19}$$

where h is the depth of the bending member in millimetres. k_{crit} may be determined from:

$$k_{crit} = \begin{cases} 1.0 & \text{for} \quad \lambda_{rel,m} \leq 0.75 \\ 1.56 - 0.75\,\lambda_{rel,m} & \text{for} \quad 0.75 \leq \lambda_{rel,m} \leq 1.4 \\ \dfrac{1}{\lambda^2_{rel,m}} & \text{for} \quad \lambda_{rel,m} > 1.4 \end{cases} \tag{20}$$

The relative slenderness for bending is:

$$\lambda_{rel,m} = \sqrt{\frac{f_{mk}}{\sigma_{m,crit}}} \tag{21}$$

where $\sigma_{m,\,crit}$ is the critical bending stress. For softwood with a solid rectangular cross-section, $\sigma m,\,crit$ is given by:

$$\sigma_{m,crit} \approx \frac{0,78b^2}{hL_{ef}} E_{0,05} \tag{22}$$

where b is the width of the beam, h is the depth of the beam, $E_{0.05}$ is the fifth percentile value of the modulus of elasticity parallel to the grain, and L_{ef} is the effective length of the beam. L_{ef} has here been taken to be:

$$L_{ef} = 0.9L + 2h \tag{23}$$

as the load is applied at the compression edge of the beam. Now, with $Wy = bh^2/6$, where Wy is the section modulus about the strong axis y, $\sigma m,\ d = M_{Ed}/W_y$, where M_{Ed} is the design moment, $k_{mod} = 0.8$ (the medium duration of loading at Service Class 1 for glulam timber), and $\gamma M = 1.25$, the following equation can be derived to check the capacity of bending moment in the ultimate limit state:

$$\frac{M_{Ed}}{M_{Rd}} = 9.375 \frac{M_{Ed}}{bh^2 f_{m,k}\, k_h\, k_{crit}} \leq 1.0 \tag{24}$$

Equation (24) represents also the utilization ratio of the bending capacity of the timber beam. The percentage reduction in moment capacity for the timber beam can now be calculated as:

$$\Delta M = (1 - k_{crit})100 \tag{25}$$

3.3.2 HEA beam. The beam should be verified with respect to its bending moment. For Classes 1 and 2 cross-sections, in which the standard HEA profiles are included, the design plastic resistance moment of the gross section, M_{pl}, must satisfy (EN 1993 Eurocode 3, 2005):

$$M_{Ed} \leq M_{pl,Rd} = \frac{W_{pl} f_y}{\gamma_{M0}} \tag{26}$$

where W_{pl} is the plastic section modulus and γ_{M0} is a material factor (=1.0). Equation (26) may be rewritten as:

$$\frac{M_{Ed}}{M_{Rd}} = \frac{\gamma_{M0} M_{Ed}}{W_{pl} f_y} \leq 1.0 \tag{27}$$

to yield the moment check without considering the lateral-torsional buckling.

Lateral-torsional buckling check

The beam should be verified with respect to its lateral-torsional buckling resistance as follows:

$$\frac{M_{Ed}}{M_{b,Rd}} \leq 1.0 \tag{28}$$

where

$$M_{b,Rd} = \chi_{LT} \frac{W_y f_y}{\gamma_{M1}} \tag{29}$$

where M_{Ed} is the design value of the moment, $M_{b,Rd}$ is the design buckling resistance moment, $W_y = W_{pl}$ for Class 1 and 2 cross-sections, and χ_{LT} is the reduction factor for lateral-torsional buckling.

Reduction factor for lateral-torsional buckling

The values of the reduction factor χ_{LT} are given by:

$$\chi_{LT} = \frac{1}{\varnothing_{LT} + \sqrt{\varnothing_{LT}{}^2 - \beta \overline{\lambda}_{LT}^{2}}} \leq \begin{cases} 1.0 \\ \dfrac{1}{\overline{\lambda}_{LT}{}^2} \end{cases} \tag{30}$$

$$\varnothing_{LT} = 0.5\left(1 + \alpha_{LT}\left(\overline{\lambda}_{LT} - \overline{\lambda}_{LT,0}\right) + \beta \overline{\lambda}_{LT}^{2}\right) \tag{31}$$

$$\overline{\lambda}_{LT} = \sqrt{\frac{f_y}{\sigma_{cr}}} = \sqrt{\frac{W_y f_y}{M_{cr}}} \tag{32}$$

where $_{LT}$ is the value of the non-dimensional slenderness for lateral-torsional buckling, M_{cr} is the elastic critical moment for lateral-torsional buckling, and α_{LT} is an imperfection factor for lateral-torsional buckling. For the case shown in Figure 1, β = 0.75 and $\lambda_{LT.0}$ = 0.4 (EN 1993 Eurocode 3, 2005), and f_y is the steel yield strength (MPa). Imperfection factors, α_{LT}, are given in Table IV; see also Table V.

The profiles of beams considered in this study are rolled I-sections with standard HEA profiles, typically used in steel constructions. Tibnor (2007) states that $h/b < 2$ for HEA profiles HEA100 to HEA600, which implies in this case that buckling curve **b** and α_{LT} = 0.34 should be used. For HEA profiles HEA600 to HEA1000, $h/b > 2$, which implies in this case that buckling curve **c** and α_{LT} = 0.49 should be used.

The elastic critical moment, M_{cr}, is based on gross cross-sectional properties and is given by (Vejlkovic *et al.*, 2015):

$$M_{cr} = C_1 \frac{\pi^2 EI_z}{(k_z L)^2} \left\{ \left[\left(\frac{k_z}{k_w} \right)^2 \frac{I_w}{I_z} + \frac{(k_z L)^2 GI_T}{\pi^2 EI_z} + (C_2 z_g)^2 \right]^{0.5} - (C_2 z_g) \right\} \tag{33}$$

where E is the Young's modulus (210 GPa), G is the shear modulus (80,770 GPa), I_z is the second moment of area about the weak axis, I_t is the torsion constant, I_w is the warping constant, L is the beam length between points that are subject to lateral restraint and z_g is the distance between the point of load application and the shear centre. Assuming that the load is applied to the top flange and acts towards the shear centre from the point of application:

$$z_g = z_a - z_s = h - \frac{h}{2} = \frac{h}{2} \tag{34}$$

where z_a and z_s are the coordinates of the point of application of the load and of the shear centre, relative to the centroid of the cross section. For doubly symmetric sections, the shear centre coincides with the centroid. It is, of course, possible to set z_g = 0, but in this case, a higher value of χ_{LT} will be obtained. The factor k_z refers to end rotation on plane. It is equivalent to the ratio of the buckling length to the system length for a compression member and is here taken to be 1.0. The factor k_w refers to end warping and is here taken to be 1.0, as special provision for warping fixity has been made (EN 1993 Eurocode 3, 2005). Note that in the common case of normal support conditions at the ends (fork supports), k_z and k_w are equal to 1.0. C_1 and C_2 are coefficients that depend on the loading and end restraint conditions. In the case considered here of a simply supported beam with equally distributed load, and with $k_z = 1$, $C_1 = 1.12$ and $C_2 = 0.45$ (Vejlkovic *et al.*, 2015).

The critical moment (kNm) can now be simplified to:

Buckling curve	a	b	c	d
α_{LT}	0.21	0.34	0.49	0.76

Source: EN 1993 Eurocode 3 (2005)

Table IV. Imperfection factors

Section Shape	Limits	Buckling curve
Rolled I-sections	h/b $\leq$ 2	b
	h/b > 2	c
Welded I-sections	h/b $\leq$ 2	c
	h/b > 2	d

Source: EN 1993 Eurocode 3 (2005)

Table V. Recommendations for the selection of lateral-torsional buckling curve for different cross-sections

$$M_{cr} = 1.12\frac{\pi^2 EI_z}{L^2}\left\{\left[\frac{I_w}{I_z}+\frac{L^2 GI_T}{\pi^2 EI_z}+(0.227h)^2\right]^{0,5}-0.227h\right\} \tag{35}$$

The equation used to check the moment can now be reformulated as:

$$\frac{M_{Ed}}{M_{Rd}} = \frac{\gamma_{M0}M_{Ed}}{\chi_{LT}W_{pl}f_y} \leq 1.0 \tag{36}$$

Equation (36) represents also the utilization ratio of the bending capacity of the steel beam. The percentage reduction in moment capacity for the steel beam can now be calculated as:

$$\Delta M = (1-\chi_{LT})100 \tag{37}$$

3.4 Ultimate limit state, shear check

3.4.1 Glued laminated timber beam. For shear with a stress component parallel to the grain, the following expression is used to check the shear capacity (EN 1993 Eurocode 5, 2004):

$$\tau_d = 1.5\frac{V_{Ed}}{b_{ef}h} \leq f_{vd} \tag{38}$$

where τd is the design shear stress, $f_{v,d}$ is the design shear strength for the relevant condition, b_{ef} is the effective width of the beam, $b_{ef} = k_{cr}\, b$, where $k_{cr} = 0.67$ for a beam assumed to be exposed to rain and solar radiation, V_{Ed} is the design shear force acting on the beam, which may be reduced near the supports by the quantity: $\Delta V_{Ed} = q_{Ed}\, h$. The design shear strength is given by:

$$f_{vd} = \frac{k_{mod}f_{vk}}{\gamma_M} \tag{39}$$

where $f_{v,k}$ is the characteristic shear strength. With the help of equations (38) and (39), the load-carrying capacity due to shearing (utilization degree) can be calculated from:

$$\frac{\tau_d}{f_{vd}} = \frac{1.875(V_{Ed}-q_{Ed}h)}{0.536bhf_{vk}} \leq 1.0 \tag{40}$$

Equation (40) represents also the utilization ratio of the shear capacity of the timber beam.

3.4.2 HEA beam. The design value of the shear force (V_{Ed}) at each cross-section is compared with the plastic shear resistance, $V_{pl.Rd}$, of the shear area (A_v) (EN 1993 Eurocode 3, 2005):

$$V_{Ed} \leq V_{Rd} \tag{41}$$

$$V_{pl,Rd} = \frac{A_v f_y}{\gamma_{M0}\sqrt{3}} \tag{42}$$

where f_y is the yield strength, $\gamma_{M0} = 1.0$ is the partial factor for the resistance of cross-sections, and A_v is the shear area, calculated here as follows:

$$A_v = A - 2bt_f + (t_w + 2r)t_f \geq \eta h_w t_w \tag{43}$$

where A is the cross-sectional area, b is the overall breadth, h is the overall depth, h_w is the depth of the web, r is the root radius, t_f is the flange thickness, t_w is the Web thickness, and $\eta = 1.0$. By combining equations (41) and (43), the following equation can be derived to check the shear force:

$$\frac{V_{Ed}}{V_{Rd}} = \frac{\sqrt{3}\gamma_{M0} V_{Ed}}{A_v f_y} \leq 1.0 \tag{44}$$

Equation (36) represents also the utilization ratio of the shear capacity of the steel beam.

4. Economy

Two different costs for each beam type have been used in the calculations presented here. The first cost is the material cost, which covers the costs incurred before the material is delivered to the construction site. The second cost is the "ready-to-assemble" cost, which includes the cost of assembly and all the attachment/connection details required to complete the construction.

Table VI presents the estimated costs on the Swedish construction market and used in this study. The prices specified are estimates, as it is difficult to determine accurate prices for building materials. A number of factors have a significant impact on the final price, such as the distance from the manufacturer to the construction site, the amount of materials required, labour costs and the current economic situation. Of course, the final costs for the two materials differ from one country to another, and the values given in Table VI are limited to Sweden. (They may also be applicable to a certain extent in other Scandinavian countries).

5. Environmental considerations

5.1 Carbon dioxide equivalent

The present study has considered the impact on the environment of only greenhouse gases. Carbon dioxide is one of the greenhouse gases and is often used as a reference value for environmentally related analyses. A measure known as the "carbon dioxide equivalent"

Cost type	Steel [SEK/kg]	Glulam [SEK/m^3]
Material cost	16.50	38.00
Cost of "ready-to-assemble"	8000.00	11000.00

Note: that due to different units for the selling prices in the Swedish construction market, the costs units are not the same for the glulam and HEA beams

Table VI.
Input cost values

(CO_{2e}) has been used in the work presented here as a reference value for comparisons. CO_{2e} is used to standardise the climate effects of various greenhouse gases.

Human-caused greenhouse gas includes carbon dioxide (CO_2), methane (CH_4) and nitrous oxide (N_2O). These gases do not have the same impact on the greenhouse effect and stay in the atmosphere for different amounts of time. To make it possible to compare the various greenhouse gases with one another, it is defined (Parry *et al.*, 2007) an index called "global warming potential" (GWP). This index indicates the warming effect of a certain amount of a greenhouse gas over a defined period of time (typically 100 years) in comparison to CO_2. Subsequently, greenhouse gases will be calculated as CO_2 equivalents, abbreviated as $CO2_e$.

The Swedish Society for Nature Conservation (Naturskyddsföreningen, 2017) states that greenhouse gases contribute to the greenhouse effect by absorbing certain wavelengths of the heat radiation that would otherwise leave the soil and then re-emit the heat to the ground.

The natural greenhouse effect is necessary for life on Earth as the climate would be about 30°C colder than it is today in the absence of this effect. However, problems arise if the concentrations of greenhouse gases increase in the atmosphere in a way that causes severe climate change that cannot be controlled. The higher concentration of carbon dioxide in the atmosphere causes a temperature increase that can eventually cause the climate to change.

5.2 Carbon storage

Wooden building components can capture CO_2 from the atmosphere and store it in their structures. This long-term process, known as "carbon sequestration", involves carbon capture and the long-term storage of atmospheric carbon dioxide (Sedjo and Sohngen, 2012). The largest potential for storing carbon is possessed by external walls, intermediate floors and roof structures that are exposed to sufficient atmospheric air. In this way, timber products may be used to reduce the effect of the climate change that results from the atmospheric accumulation of greenhouse gases, which are naturally released by burning fossil fuels. Consequently, carbon sequestration can have a positive impact on the environment, as it involves removing carbon dioxide from the atmosphere, and creating in this way what are known as "negative emissions". The advantages of using timber materials to store atmospheric carbon in the built environment have received considerable attention in the scientific literature (Kutnar and Hill, 2017).

5.3 Input values

Table VII shows the emission and sequestration of carbon dioxide (kg CO_{2e}/kg) to and from the air from timber and steel materials (Hill and Dibdiakova, 2016). These values reflect emissions from the complete life cycle of the construction material. The values depend on the production process. In the work presented here, the mean values of values from several production processes have been calculated. The amount of CO_{2e} for the glulam beam can now be calculated as:

$$CO_{2e} = XbhL\rho_{mean}\,(\text{kg}) \tag{45}$$

where ρ_{mean} is the mean density of the glulam beam (here 480 kg/m^3), L is the beam span and X is the mean value of sequestration or emission (X = 0.43 kg CO_{2e}/kg if the carbon storage is excluded and $X = -1.3$ kg CO_{2e}/kg if it is included). For the steel beam:

Material	Input value (kg CO_{2e}/kg)	Comment
Steel		
	0.5	
	0.6	
	0.7	
Glulam timber	0.6	Exc. carbon storage
	0.5	Exc. carbon storage
	0.4	Exc. carbon storage
	0.2	Exc. carbon storage
	−1	Incl. carbon storage
	−1.1	Incl. carbon storage
	−1.2	Incl. carbon storage
	−1.4	Incl. carbon storage
	−1.6	Incl. carbon storage

Source: Hill and Dibdiakova (2016)

Table VII. The emission of carbon dioxide to the air from timber and steel materials, based on reference

$$CO_{2e} = 0.6g\,L\,(\mathrm{kg}) \tag{46}$$

where g is the weight of the beam per meter (kg/m).

6. Results and discussion

6.1 Structural design

Table I presents the beam dimensions (cross-sections and span lengths) of the beams considered here. Timber strength class GL30c and HEA steel sections (rolled I-sections) were chosen, as these are commonly used in practice.

The characteristic strength and stiffness properties and densities for the strength class GL30c were obtained from Swedish Wood (2016). The corresponding sources for values of the steel sections were Tibnor (2007) and EN 1993 Eurocode 3 (2005). The partial safety factor, γ_d, was taken to be 1.0.

The dimensions were chosen such that the load-bearing capacity was reasonable. Tables II and III show the results of structural calculations carried out in Section 3. The moment capacities check (utilization ratios of the bending capacities) of the two beam types are almost the same. Consequently, as the beam span increases, the load-bearing capacity will require that greater dimensions are used to safely bear the loading actions. Moreover, the control of shear capacity (utilization ratios of the shear capacities) for the two beams differs. The shear capacity of the glulam beams is reached at a lower cross section than is the case for the steel beams, which shows that steel beams resist shearing better than timber beams.

The resistance to lateral-torsional buckling of the timber beams is higher than that of the steel beams. As span length increases, the lateral torsional buckling increases remarkably faster for steel beams than for timber beams. Lateral-torsional buckling does not occur for timber beams shorter than 8 m. Thus, the steel beams studied are more prone to lateral torsional buckling than the glulam beams, and lateral torsional buckling will be the decisive criterion for the steel beams with respect to moment capacity. In practice, this implies that lateral restraints, or staggering, are more important for steel beams than for timber beams, especially at large beam spans. Evidently, this has economic consequences.

Timber beams are more sensitive to deflection as the beam span increases, in contrast to the case of steel beams. Lateral supports or restraints can be used to decrease the deflection, or glulam beams can be produced with a camber, to bend or curve upwards at the centre. Once again, this may lead to additional costs.

The cross-sectional area of the beams must also be considered. The dimensions of the glulam beams are considerably larger than the dimensions of the steel beams for the same span. In practice, it can be difficult to justify using a timber beam with a height of 1305 mm instead of a steel beam with a height of 690 mm (HEA700). In some cases, it may not even be possible to use glulam beams due to their large dimensions. The construction engineer or project manager must decide which beam type is to be used, considering the space available, as several methods are available to tackle this issue.

Note that the work presented here has included an extreme fall of snow action, which is only found in heavily snowed areas such as the northern parts of Sweden.

6.2 Economy

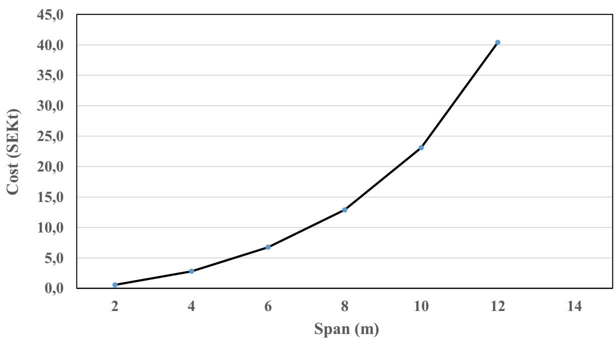
Costs have been calculated based on the selected dimensions (Table I) for the two beam types. Figures 2 and 3 show the estimated material costs for the two beam types, and Figures 4 and 5 show the estimated costs for the two beam types ready to be assembled in the building structure.

Glulam beams are always cheaper than steel beams of HEA profile. The material costs do not differ greatly: the big difference is the cost of "ready-to-assemble".

Analysis of the input values shows that labour costs and costs for connection fittings are significant, as are peripheral costs (such as the mounting of steel beams):

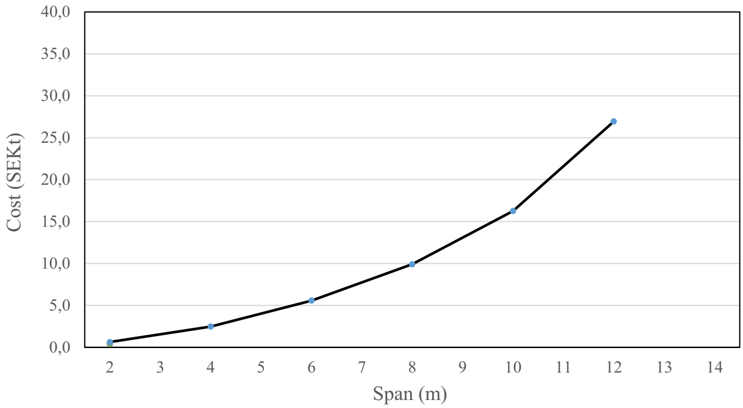
$$\text{Glulam}: \frac{\text{Materal cost}}{\text{Cost of (ready} - \text{to} - \text{assemble)}} \cdot 100 = \frac{8,000\text{kr/m}^3}{11,000\text{kr/m}^3} \cdot 100 \approx 72.7\%$$

$$\text{Steel}: \frac{\text{Materal cost}}{\text{Cost of (ready} - \text{to} - \text{assemble)}} \cdot 100 = \frac{16.50\text{kr/kg}}{38.00\text{kr/kg}} \cdot 100 \approx 43.4\%$$



Note: SEKt is thousand Swedish kroner

Figure 2.
Material costs for steel beams



Note: SEKt is thousand Swedish kroner

Figure 3.
Material cost estimate for glulam beam

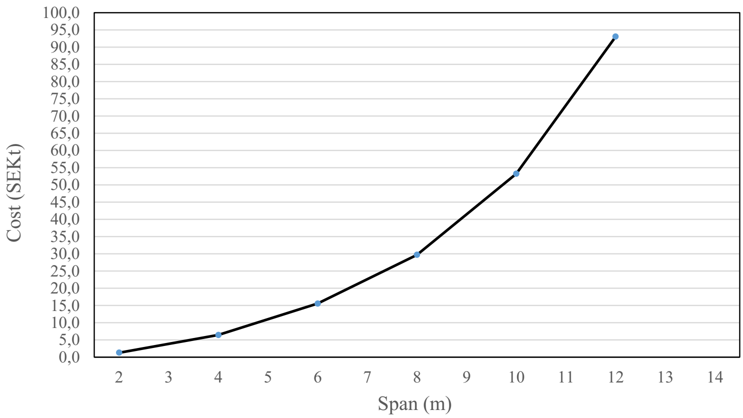


Figure 4.
Cost estimates of "ready-to-assemble" for steel beams

The material cost of the glulam beams is 72.7 per cent of the total, while for the steel beams, it is 43.4 per cent.

It must also be remembered that the market prices of steel products vary greatly with time. This makes it difficult to calculate the material cost of steel structures. It is, however, uncertain how important this is in the context, as the work presented earlier in the text shows that material costs are not the largest cost item for steel beams.

The analysis of structural design has shown that steel beam with HEA profile require more staggering (or lateral restraints) than glulam beams. This will incur additional costs, which must be considered.

In many cases, it is preferable to select cross-sectional dimensions for glulam beams that provide the lowest area. This can give a smaller total volume of material, and thereby lower material costs.

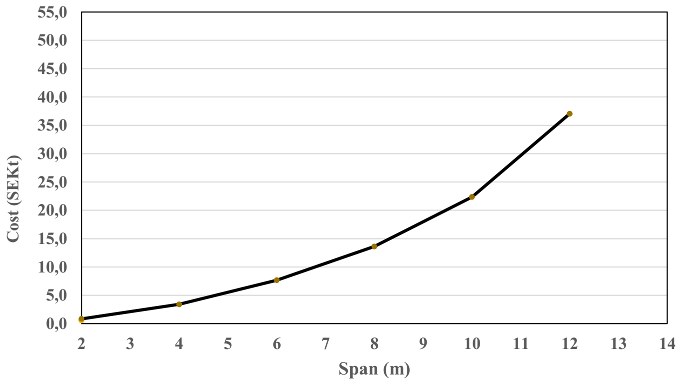


Figure 5.
Cost estimates for "ready-to-assemble" for glulam beams

6.3 Environmental considerations

Section 4 has described the method for calculating environmental impact, the results of which are given in Tables VIII and IX. Note that negative values indicate carbon sequestration.

Emission of carbon dioxide from the glulam beams is lower than that from steel beams. The difference between the two beam types increases as the span increases. A reduction of approximately 50 per cent can be achieved when using glulam timber instead of steel for a beam with a span of 12 m. Glulam beams are the most environmentally friendly option, and the carbon-storage capacity of the wood ensures that using such beams has a positive impact on the greenhouse effect and therefore the environment. However, the benefits of using wood products to store atmospheric carbon dioxide can be achieved only if the products have a sufficiently long life (Kutnar and Hill, 2017; Hill and Dibdiakova, 2016). Carbon emissions into the atmosphere have a temporal aspect, and the carbon may

Table VIII.
Results of carbon dioxide equivalent, timber class GL30c

Span (m)	$b \times h$, GL30c	kg CO_{2e} (without carbon storage)	kg CO_{2e} (with carbon storage)
2	215 × 180	16.0	−48.3
4	215 × 360	63.9	−193.2
6	215 × 540	143.8	−434.7
8	215 × 720	255.6	−772.8
10	215 × 945	419.4	−1267.8
12	215 × 1305	694.9	−2100.9

Table IX.
Results of carbon dioxide equivalents, steel beam

Span	Steel profile	kg CO_{2e}
2	HEA100	20.0
4	HEA200	101.5
6	HEA260	245.5
8	HEA320	468.5
10	HEA450	840.0
12	HEA700	1468.8

eventually be returned to the atmosphere (Ekvall, 2006). This means that the storage of carbon is insignificant. The use of wood products as a carbon buffer suffers from a further problem: when the house is demolished and burned, the carbon stored in the wood products will be returned to the atmosphere (Gustavsson *et al.*, 2006).

The ability of wood to store carbon is essential if the use of wood products increases over time (Sandberg *et al.*, 2001). In the case of roof beams and building elements in general, more wooden houses must be built than are demolished. Moreover, the carbon-storage capacity will become more relevant if forests are managed in an efficient way. The advantages of using wood in construction will simply not be realised if the amount of forests is greatly reduced.

6.3.1 Other comparison aspects. The residual waste from the different materials can be treated differently. It may be that wood is the better option (Hill and Dibdiakova, 2016), but the differences are small. Wood has a much smaller environmental impact if it is recycled (Sandberg *et al.*, 2001) because in this case, the absorbed carbon is not returned to the atmosphere. However, in some cases, wood is less likely to be recycled and more likely to be burned. It remains true, however, that wood is the better option from an environmental perspective, even when the wood is burned, when compared with steel (which is recycled), which is the most likely scenario (Sandberg *et al.*, 2001; Petersen and Solberg, 2002).

7. Concluding remarks

We have shown that using glulam beams in construction has environmental and economic benefits greater than those of using steel beams. Glulam beams can have a positive effect on the environment since the wood has a carbon-storage capacity. The economic benefits of building with glued laminated wood are considerable, where the ready-to-assemble costs are the major difference in cost between the two types of beam. The cost analysis has been based on prices in Sweden and can be generalized to Scandinavian countries and other similar countries that are rich in forests.

Glulam beams have a lower risk of lateral torsional buckling than steel beams have. One disadvantage of using glulam beams are the large dimensions that may be required, especially at longer spans. If this is a problem, steel beams are the better option. Glulam beams with smaller widths should be selected for use over longer spans. In this way, the problem of the reduction in available space caused by the use of deep beams can be minimized. A wider beam is not significantly more expensive than a narrower beam over a given span. In contrast, the cross-sectional dimensions of steel beams cannot be selected freely since these are predetermined. If the design criteria are not satisfied, a larger profile size or another profile must be selected, or a beam built up as welded steel sections must be used.

One of the positive aspects of using timber in construction engineering is the environmental benefits that this brings. The dry mass of wood is 50 per cent carbon, and this carbon is removed from the atmosphere, which mitigates the climate change caused by the greenhouse effect. If more buildings with wooden frames are built, the amount of wood products used will increase with time, and the carbon-storage capacity of wood will be increase. The net storage of carbon dioxide that would occur would have a direct positive impact on the greenhouse effect and hence the environment. If more buildings with steel beams are built, this process will not occur. However, the environmental impact of steel beams is much lower if the steel is recovered after demolition.

Sweden's major assets of vast areas of forest and a long tradition of sustainable forestry provide the prerequisites for reaping the benefits of using wood material in new construction. This does not apply to all countries. In many parts of the world, forests are not as plentiful as they are in Sweden. It remains to be determined whether the use of wood in construction can be justified in countries in which forests and wood materials are not abundant resources.

As mentioned earlier, it is shown that using glulam beams in construction has economic benefits greater than those of using steel beams. However, it is important to understand that this conclusion is based on the present Swedish market prices. In general, the prices can vary due to the economic factors, which affect a business or an investment's value. In the long run, various economic factors need to be taken into account when determining the expected future value of steel and timber elements. Key economic factors include labour costs, interest rates, government policy, inflation rate, taxes and management.

Finally, sustainable buildings can practically be achieved by studying the environmental and economic values of different building elements. By integrating these two dimensions with the technical design of building elements made of steel and/or timber, a sustainable built environment can be idealised in the long run.

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